

```

/*****
* AUTHOR:      CYRIL ANDRE B. DURANGO
* CLASS:       CPE 3101L - INTRODUCTION TO HDL
* SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
* -----
* PROJECT FILE TITLE: Problem_B.v
*****/

```

```

module Problem_B(
    input          clk_FPGA,
    input wire [1:0] A,
    output wire [4:0] Z
);
    wire clk;

    Clock_converter C1k1(clk_FPGA, clk);
    Problem_B_Code PBC1(clk, A, Z);
endmodule

```

```

module Problem_B_Code(
    input          clk,
    input wire [1:0] A,
    output reg [4:0] Z
);
    always @(posedge clk)
    begin
        if (A == 2'd3)
            Z <= 4'd0;

        else if (A == 0) begin
            Z <= Z + 4'd2;
            if (Z == 30)
                Z <= Z;
        end

        else if (A == 1) begin
            Z <= Z - 4'd2;
            if (Z == 0)
                Z <= Z;
        end

        else if (A == 2)
            Z <= Z;
    end
endmodule

module Clock_converter(
    input wire clk_FPGA,
    output reg clk_converted
);
    reg [25:0] counter;

    always @(posedge clk_FPGA)
    begin
        counter <= counter + 1;
        if (counter == 25'd25000000) begin
            clk_converted <= ~clk_converted;
            counter <= 0;
        end
    end
endmodule

```

Quartus Prime Lite Edition - Z:/Finals/Problem_B/Problem_B - Problem_B

File Edit View Project Assignments Processing Tools Window Help

Project Navigator Hierarchy Entity Instance

MAX 10: 10M50DAF484C7G

Problem_B

Table of Contents

- Flow Summary
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Flow Summary

Flow Status: Successful - Fri Dec 13 09:31:31 2024

Quartus Prime Version: 20.1.1 Build 720 11/11/2020 SJ Lite Edition

Revision Name: Problem_B

Top-level Entity Name: Problem_B

Family: MAX 10

Device: 10M50DAF484C7G

Timing Models: Final

Total logic elements: 64 / 49,760 (< 1 %)

Total registers: 31

Total pins: 8 / 360 (2 %)

Total virtual pins: 0

Total memory bits: 0 / 1,677,312 (0 %)

Embedded Multiplier 9-bit elements: 0 / 288 (0 %)

Total PLLs: 0 / 4 (0 %)

UFM blocks: 0 / 1 (0 %)

ADC blocks: 0 / 2 (0 %)

IP Catalog

Installed IP

No Selection Available

Project Directory

Library

- Basic Functions
- DSP
- Interface Protocols
- Memory Interfaces and Controllers
- Processors and Peripherals
- University Program

Search for Partner IP

Tasks

Compilation

Task

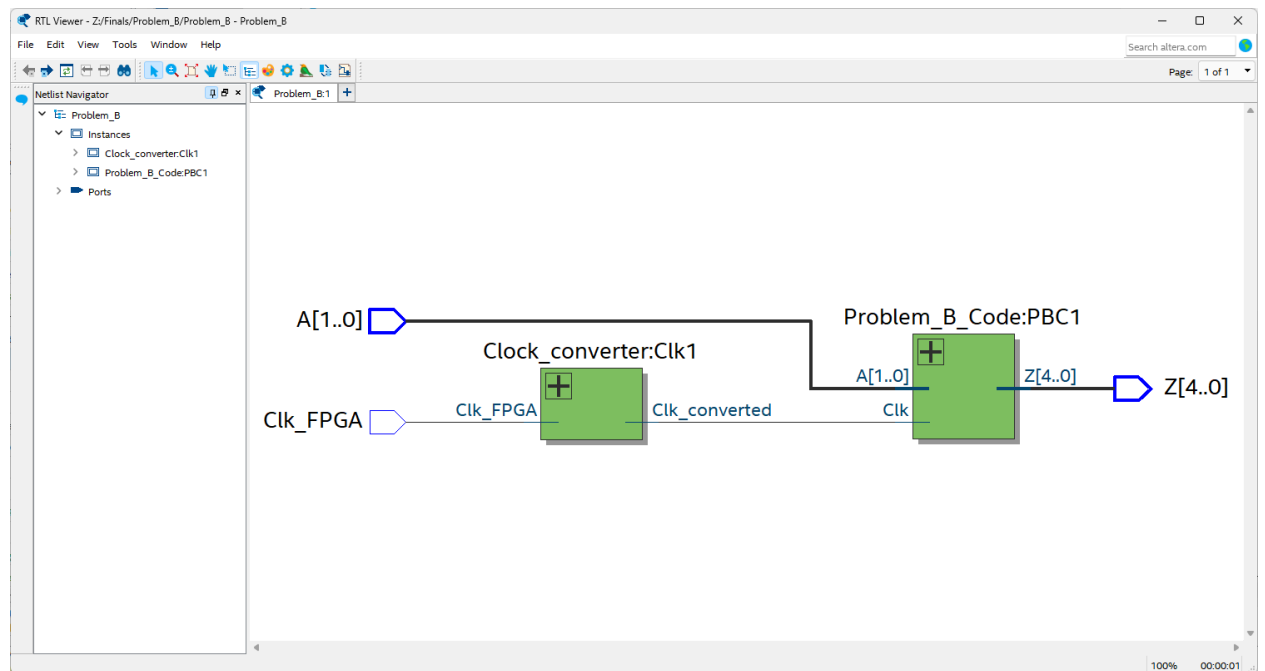
- Compile Design
- Analysis & Synthesis
 - Edit Settings
 - View Report
 - Analysis & Elaboration
 - Partition Merge
 - Netlist Viewers
 - Design Assistant (Post-Mapping)
 - I/O Assignment Analysis
 - View Report
 - Pin Planner
- Fitter (Place & Route)
- Assembler (Generate programming file)
- Timing Analysis
- EDA Netlist Writer
- Edit Settings
- Program Device (Open Programmer)

Messages

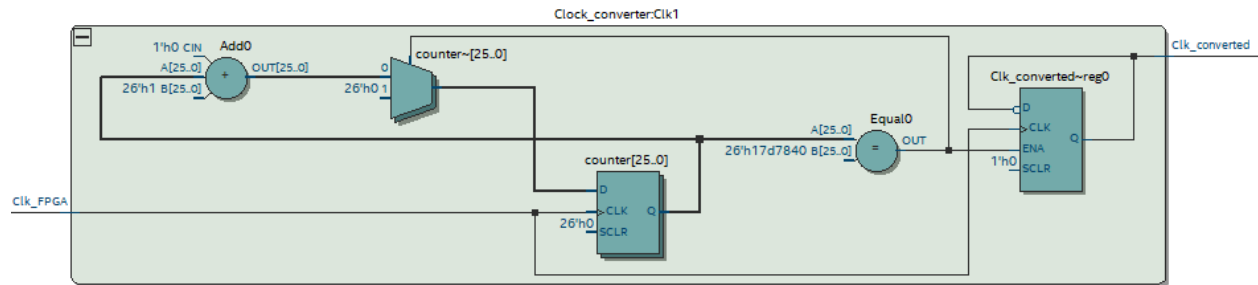
System (8) Processing (121)

100% 00:00:28

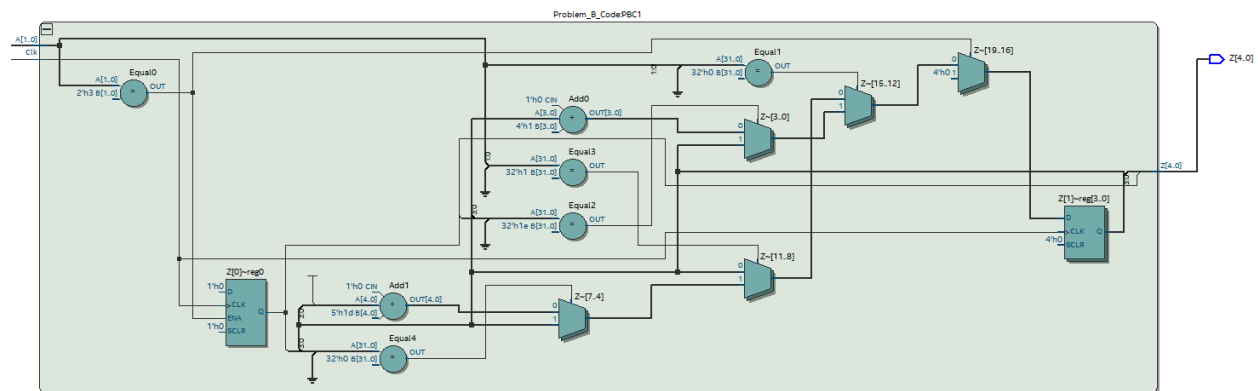
RTL VIEWS:



PROBLEM_B | MAIN RTL VIEW

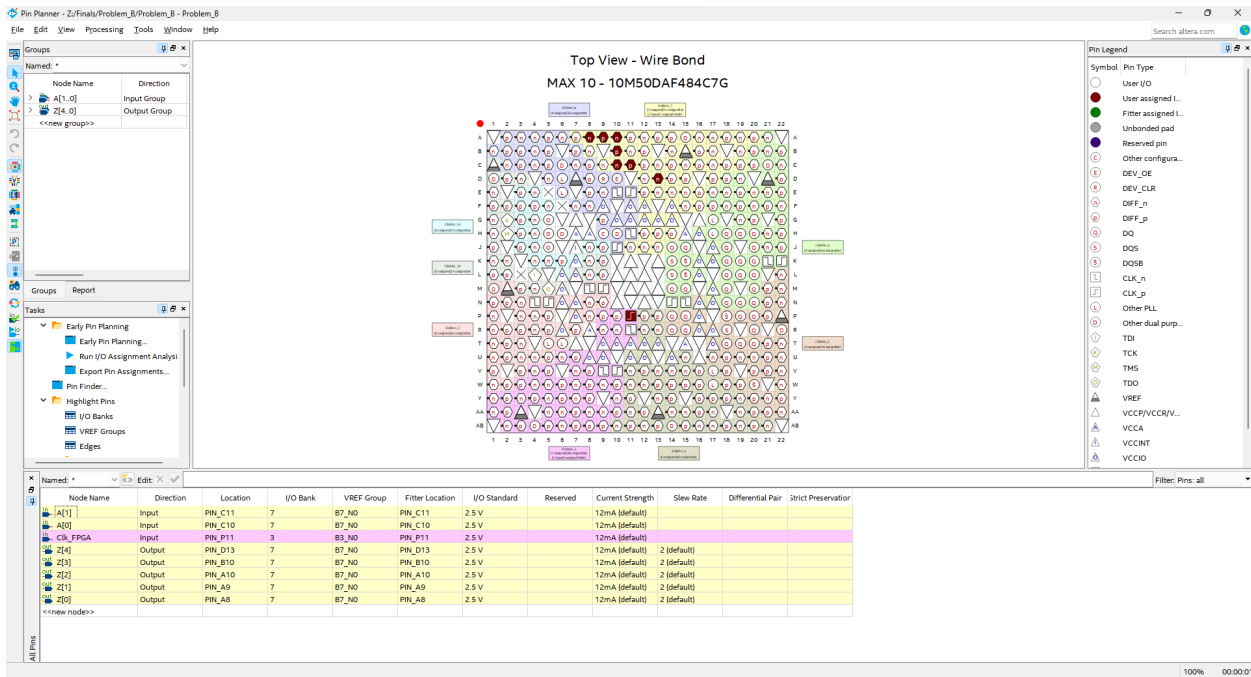


CLOCK_CONVERTER EXPANDED RTL VIEW



COUNTER EXPANDED RTL VIEW

PIN ASSIGNMENTS:



NAME IN VERILOG	FPGA PIN ASSIGNEMNT	CORRESPONDING I/O DEVICE
CLK_FPGA	PIN_P11	50 MHZ Clock Input
A[0]	PIN_C10	SW0
A[1]	PIN_C11	SW1
Z[0]	PIN_A8	LEDR0
Z[1]	PIN_A9	LEDR1
Z[2]	PIN_A10	LEDR2
Z[3]	PIN_B10	LEDR3
Z[4]	PIN_D13	LEDR4